INDIVIDUAL PROJECT - MILESTONE 2 SUBMISSION

CLIENT IMPLEMENTATION REVIEW

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Module: CMPG345

Project: North-West Electrical Engineering Consultants

Network Design and Implementation

Assigned Feature: ACLs (Access Control Lists)

1. INTRODUCTION

This document presents the implementation of the network design for North-West Electrical Engineering Consultants, an engineering company based in Klerksdorp that requires a reliable, secure, and scalable network.

For Milestone 2, the following was completed:

. Full network build in Cisco Packet Tracer
. VLAN configuration across all departments
. Inter-VLAN routing on a Layer 3 core switch
. Implementation of Access Control Lists (ACLs) as the assigned feature
. SSH remote management restricted to a single off-site administrator
. Comprehensive testing to verify connectivity and security

The network was built according to the IP addressing plan using the assigned block 192.168.34.0/24, divided into logical subnets for each department.

2. NETWORK IMPLEMENTATION

2.1 VLAN Configuration

The network was divided into the following VLANs:

VLAN ID	Name	Subnet	Purpose
10	Engineering	192.168.34.0/27	Engineering workstations
20	HR-Admin	192.168.34.32/27	HR and admin staff
30	Boardroom	192.168.34.64/28	Presentation devices + WAP
40	Servers	192.168.34.80/28	DHCP/File server infrastructure
50	Management	192.168.34.96/28	Management workstations

99	Native	192.168.34.128/28	Remote admin and native traffic

2.2 Inter-VLAN Routing

A Layer 3 core switch (Cisco 3560) was used to perform inter-VLAN routing. SVIs (Switch Virtual Interfaces) were configured for each VLAN to act as default gateways, and IP routing was enabled with the "ip routing" command.

2.3 Trunk Links

Trunk links (802.1q encapsulation) were configured between all Access Switches and the Core Switch, with Native VLAN 99 on all trunks.

3. ASSIGNED FEATURE: ACCESS CONTROL LISTS (ACLs)

The assigned networking feature for this project is ACLs. ACLs were implemented on the Layer 3 Core Switch to enforce the following security policies:

3.1 ACL 1: Block Engineering from accessing HR

ip access-list extended ACL_ENG_HR deny ip 192.168.34.0 0.0.0.31 192.168.34.32 0.0.0.31 permit ip any any interface vlan 10 ip access-group ACL_ENG_HR in

Purpose: Prevents Engineering staff from accessing HR department resources, protecting sensitive HR and payroll data.

3.2 ACL 2: Block Boardroom from accessing internal networks

ip access-list extended ACL_BOARD_IN deny ip 192.168.34.64 0.0.0.15 192.168.34.0 0.0.0.224 deny ip 192.168.34.64 0.0.0.15 192.168.34.32 0.0.0.224 deny ip 192.168.34.64 0.0.0.15 192.168.34.96 0.0.0.224 permit ip any any interface vlan 30 ip access-group ACL_BOARD_IN in

Purpose: Visitors using the boardroom Wi-Fi or wired ports cannot access internal departments, but they can still access the internet and the shared file server.

3.3 ACL 3: Restrict SSH to the Remote Administrator

ip access-list extended ACL_ADMIN_SSH permit tcp host 192.168.34.130 any eq 22 deny tcp any any eq 22 permit ip any any interface vlan 99 ip access-group ACL_ADMIN_SSH in

Purpose: Only the off-site administrator (192.168.34.130) is permitted to establish SSH sessions to manage network devices. All other devices are blocked.

3.4 Additional SSH Block on Engineering VLAN

ip access-list extended ACL_ENG_SSH_BLOCK deny tcp 192.168.34.0 0.0.0.31 host 192.168.34.129 eq 22 permit Ip any any interface vlan 10 ip access-group ACL_ENG_SSH_BLOCK in

Purpose: Provides a secondary layer of protection, blocking Engineering devices from even reaching the SSH service on the Core Switch.

3.5 SSH Configuration

ip domain-name project. Local username admin privilege 15 secret cisco123 line vty 0 4 access-class 10 in login local transport input ssh access-list 10 permit 192.168.34.130

Purpose: Telnet is disabled. Only SSH is allowed for remote management, and the VTY access-class restricts connections to the Remote Admin IP.

4. TESTING EVIDENCE

The following tests were performed in Cisco Packet Tracer to verify that both connectivity and security policies work as designed.

5. CONCLUSION

The Milestone 2 implementation successfully delivered a secure, segmented network for North-West Electrical Engineering Consultants. All VLANs were configured, inter-VLAN routing was enabled on the Layer 3 core switch, and ACLs were implemented to enforce the security policies specified in the project brief.

Testing confirmed that:

. Authorized traffic flows correctly (Engineering → Server, Boardroom → Server)
. Unauthorized traffic is blocked (Engineering → HR, Boardroom → HR)
. Remote management is restricted to a single authorized administrator via SSH
. Telnet has been disabled in favor of secure SSH

The network meets the client's requirements for a reliable, secure, and scalable design, and is ready for future expansion.